



Cambridge International AS & A Level

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MATHEMATICS**9709/33**

Paper 3 Pure Mathematics 3

October/November 2024**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

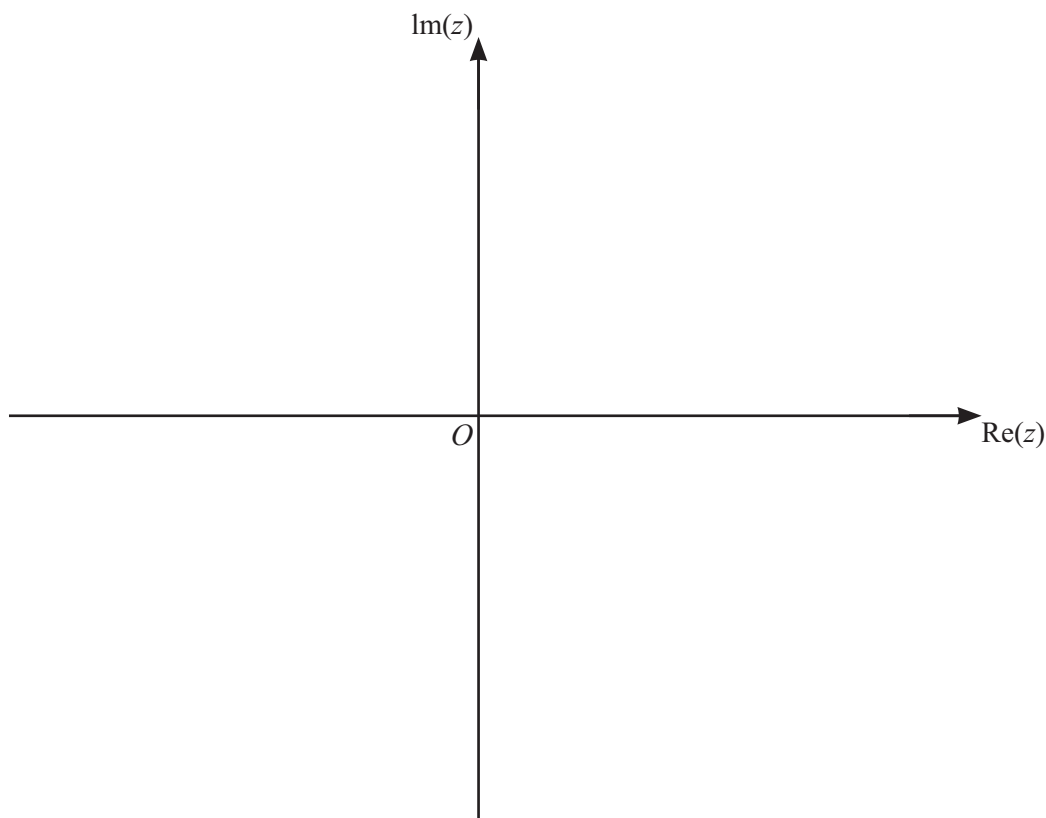
This document has **20** pages.



1 The complex number z satisfies $|z| = 2$ and $0 \leq \arg z \leq \frac{1}{4}\pi$.

(a) On the Argand diagram below, sketch the locus of the points representing z . [2]

(b) On the **same diagram**, sketch the locus of the points representing z^2 . [2]





2 Let $f(x) = 2x^3 - 5x^2 + 4$.

(a) Show that if a sequence of values given by the iterative formula

$$x_{n+1} = \sqrt{\frac{4}{5-2x_n}}$$

converges, then it converges to a root of the equation $f(x) = 0$.

[2]

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(b) The equation has a root close to 1.2 .

Use the iterative formula from part (a) and an initial value of 1.2 to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places.

[3]

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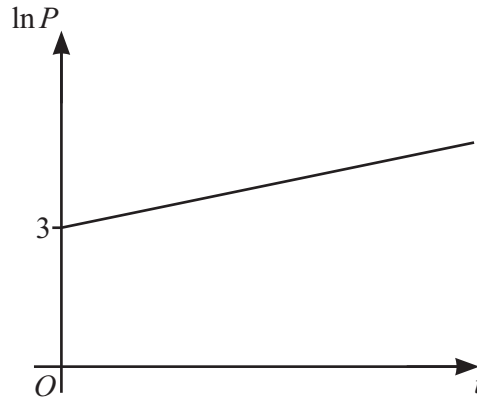
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3



The number of bacteria in a population, P , at time t hours is modelled by the equation $P = ae^{kt}$, where a and k are constants. The graph of $\ln P$ against t , shown in the diagram, has gradient $\frac{1}{20}$ and intersects the vertical axis at $(0, 3)$.

- (a) State the value of k and find the value of a correct to 2 significant figures. [3]

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- (b) Find the time taken for P to double. Give your answer correct to the nearest hour. [2]

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4 Find the complex number z satisfying the equation

$$\frac{z-3i}{z+3i} = \frac{2-9i}{5}.$$

Give your answer in the form $x + iy$, where x and y are real.

[5]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

[3]

[illegible]



[3]

[illegible]

6 The lines l and m have vector equations

$$l: \mathbf{r} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k} + \lambda(-\mathbf{i} + 2\mathbf{k}) \quad \text{and} \quad m: \mathbf{r} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k} + \mu(2\mathbf{i} - \mathbf{j} + 5\mathbf{k}).$$

Lines l and m intersect at the point P .

- (a) State the coordinates of P . [1]

[illegible]

- (b) Find the exact value of the cosine of the acute angle between l and m . [3]

[illegible]



$$x = 3 \sin 2t, \quad y = \tan t + \cot t,$$

(a) Show that $\frac{dy}{dx} = \frac{-2}{3 \sin^2 2t}$. [5]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dotted lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for writing practice. There is no text or other markings on the page.



- (b)** Find the equation of the normal to the curve at the point where $t = \frac{1}{4}\pi$. Give your answer in the form $py + qx + r = 0$, where p , q and r are integers. [3]

This image shows a full page of primary-ruled paper. It features approximately 20 horizontal dotted lines spaced evenly down the page, providing a guide for handwriting practice. The paper is otherwise blank, with no margins, text, or other markings.

(a) Express $f(x)$ in partial fractions.

[3]

[illegible]



- (b)** Hence obtain the expansion of $f(x)$ in ascending powers of x , up to and including the term in x^2 .
[4]

[illegible]

- (c) State the set of values of x for which the expansion in part (b) is valid. [1]

[illegible]

[3]

[illegible]



(b) Hence show that $\int_2^{2\sqrt{3}} \frac{x^4 + 16}{x^2 + 4} dx = \frac{4}{3}(\pi + 4)$.

[5]



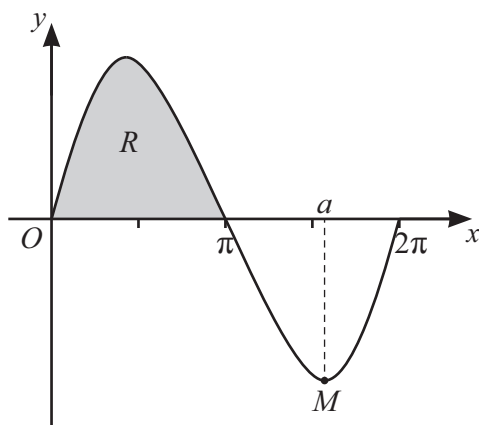
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(b) It is given that when $t = 0$, $h = 50$.

Find the time taken for the depth of water in the tank to reach 80 cm. Give your answer correct to 2 significant figures. [5]

This image shows a full page of a handwriting practice worksheet. It consists of multiple rows of horizontal dotted lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.



The diagram shows the curve $y = 2 \sin x \sqrt{2 + \cos x}$, for $0 \leq x \leq 2\pi$, and its minimum point M , where $x = a$.

- (a) Find the value of a correct to 2 decimal places. [5]

[illegible]



(b) Use the substitution $u = 2 + \cos x$ to find the exact area of the shaded region R .

[6]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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